



國立成功大學 統計學系
Department of Statistics, NCKU.

Fused Lasso

Final Project Meeting I

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12/08

Outline

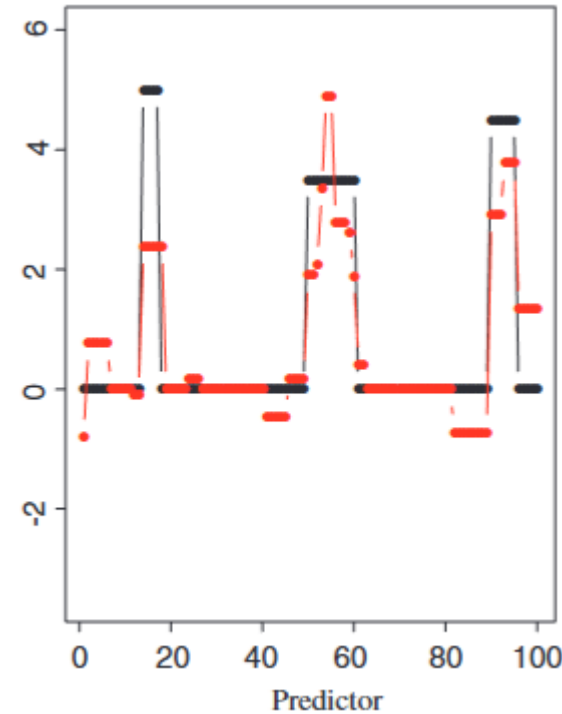
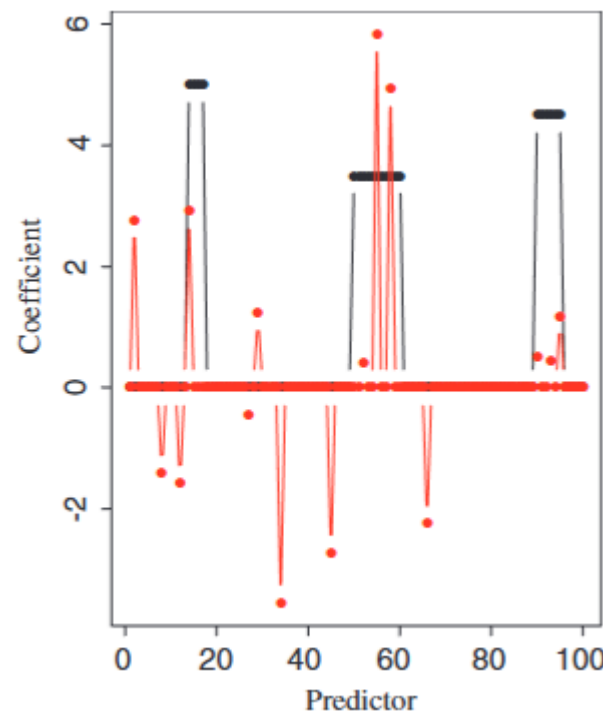
- Current Progress
- Replicate the Methodology on a Different Dataset
- Future Work

Current Progress

Replicate of Lasso and Fused Lasso simulation experiment.

$$\hat{\beta} = \operatorname{argmin}_{\beta} \sum_{i=1}^N \left(y_i - \sum_{j=1}^p x_{ij} \beta_j \right)^2$$

subject to $\sum_{j=1}^p |\beta_j| \leq s$



$$\hat{\beta} = \operatorname{argmin}_{\beta} \sum_{i=1}^N \left(y_i - \sum_{j=1}^p x_{ij} \beta_j \right)^2$$

subject to $\sum_{j=1}^p |\beta_j| \leq s_1$ and $\sum_{j=2}^p |\beta_j - \beta_{j-1}| \leq s_2$

Fig 1. Simulated example, with $p=100$ predictors having coefficients shown by the black lines:

(left) **lasso** solution (red), using $s_1 = 35.6$ and $s_2 = \infty$;

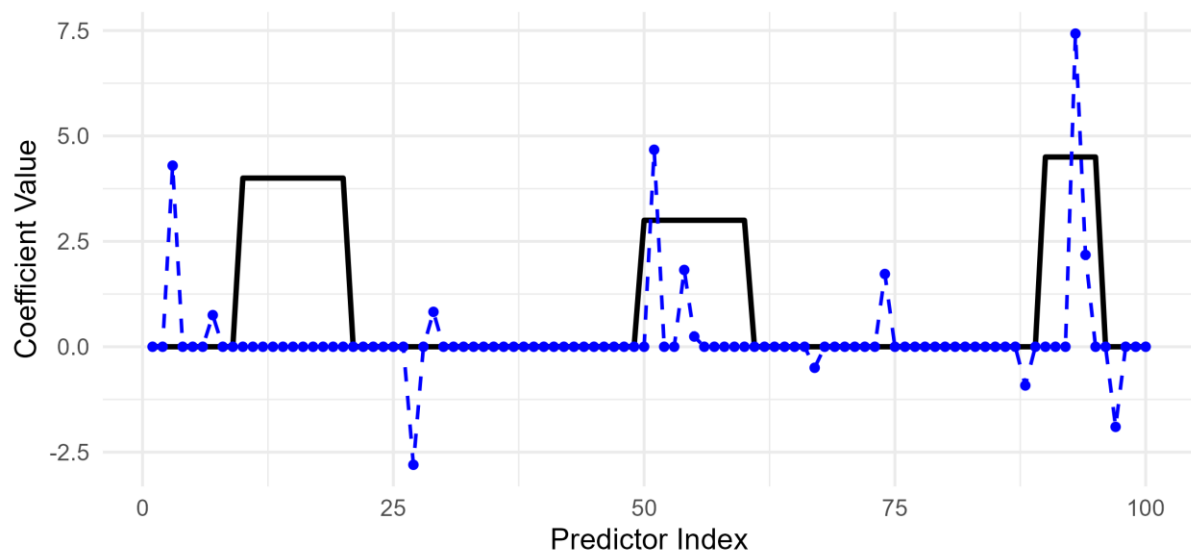
(right) the **fused lasso**, using $s_1 = \sum_{j=1}^p |\beta_j|$ and $s_2 = \sum_{j=2}^p |\beta_j - \beta_{j-1}|$, where β is the true set of coefficients.[1]

Replicate Results

Standard Lasso Simulation

$N = 20$, $p = 100$, $\sigma = 0.75$

Legend ● Lasso Estimate — True Beta



Replication of Fused Lasso Simulation

$N = 20$, $p = 100$, $\sigma = 0.75$

Legend ● Fused Lasso Estimate — True Beta

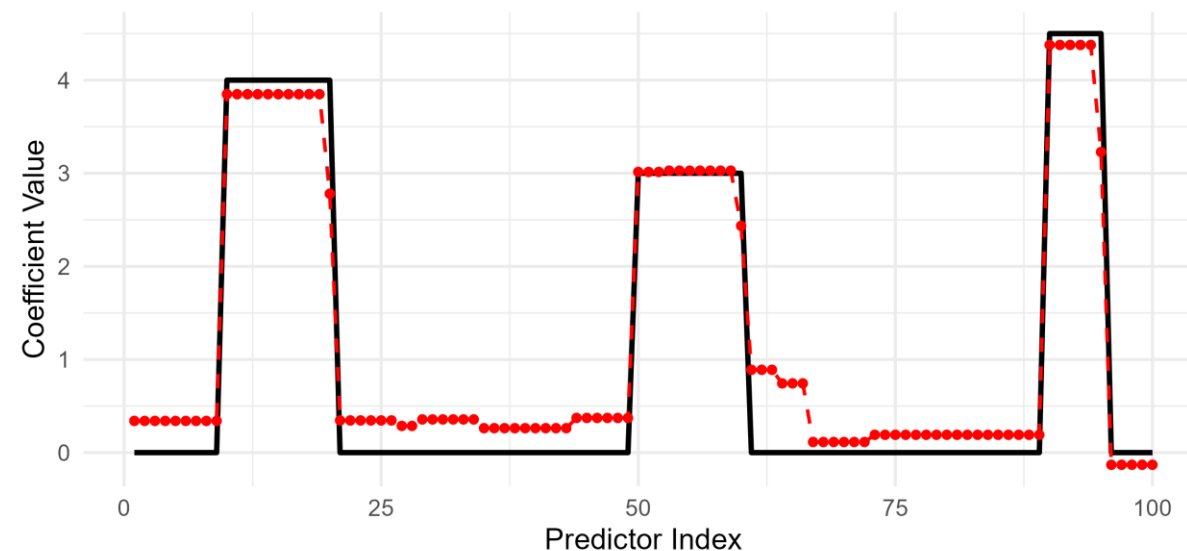


Fig 2. Replication of Simulated example, with $N=20$, $p=100$, $\sigma=0.75$ predictors having coefficients shown by the black lines: (left) Lasso (right) Fused lasso.

Replicate on Different Dataset (Store Sales)

Description: Predict sales for the thousands of product families sold at Favorita stores located in Ecuador. The training data includes dates, store and product information. [2]

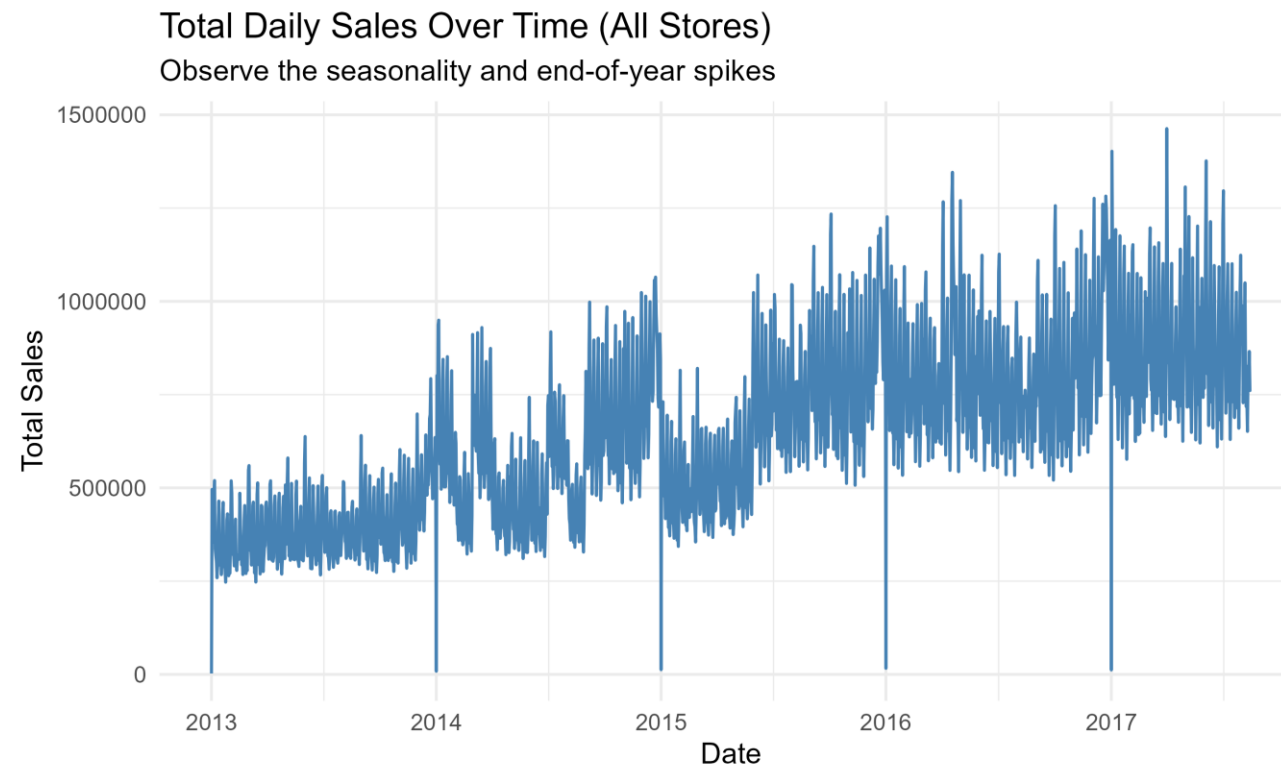


Fig 3. Total Daily Sales Over Time (2013-01-01~2017-08-15)

Replicate on Different Dataset (Store Sales)

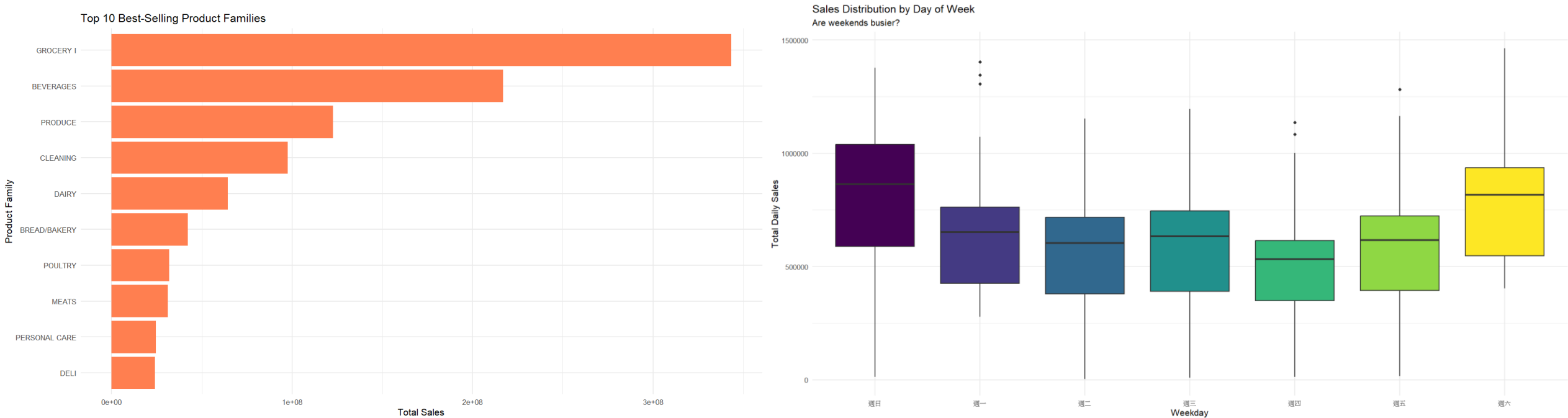


Fig 4. The left plot shows the **top 10 best-selling products**, and the right plot shows the **sales distribution by day of the week**.

Replicate on Different Dataset (Store Sales)

Comparison: Fused Lasso vs. Standard Lasso on Store Sales

Store 1, Grocery I (2016), N = 592

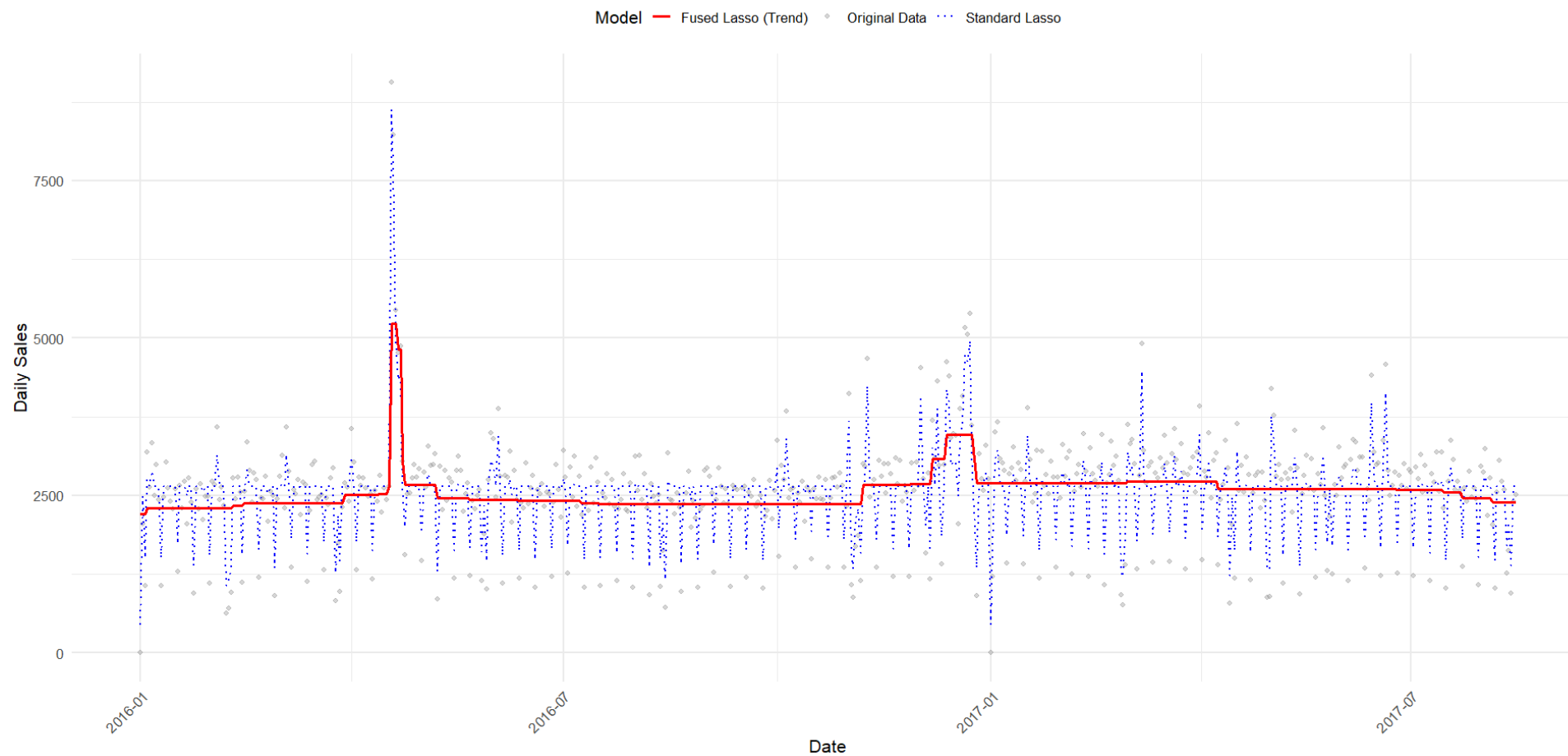


Fig 5. Total Daily Sales Over Time (All Stores)

Future Work

- Using Cross-Validation
- Seasonality Decomposition
- 2D Fused Lasso(Neighboring stores or similar products)
- Model Comparison

Thanks for your Attention

Have any Questions?